gpeyt1-04 · Atterberg / LI

GEO PE · Long Calculation Problems I — Topic 1: Site Characterization

Problem

A Shelby-tube sample of gray clay from 30 ft has LL = 58, PL = 24, and natural water content w = 52%. The torvane reading was 0.35 tsf, the pocket penetrometer 0.5 tsf, and 100% passes the No. 200 sieve. The engineer wants the liquidity index and a USCS fine-grained symbol from the plasticity chart.

Most nearly, the LI and USCS symbol are:

A. LI = 0.18, CH
B. LI = 0.82, MH
C. LI = 0.48, CL
D. LI = 0.82, CH

Given equations / data

> PI = LL − PL; LI = (w − PL)/PI; A-line: PI = 0.73(LL − 20)

Worked solution

1. PI = 58 − 24 = 34
2. LI = (52 − 24)/34 = 0.824
3. A-line at LL = 58: 0.73(58−20) = 27.7 < PI = 34 → above A-line
4. LL ≥ 50 and above A-line → CH; LI near 0.9 → soft, near-liquid consistency

Final answer

LI = 0.82, CH

Why the other values are wrong

- **LI = 0.18, CH** — computed consistency index (LL − w)/PI
- **LI = 0.82, MH** — read the soil as below the A-line
- **LI = 0.48, CL** — divided by LL and used LL < 50 boundary

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.2 Atterberg Limits, p. 132; §3.7.2 plasticity chart

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=DJaHlgyaOAg (no creator values copied).

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